

Zephyr Rollout Decision Analysis

Timing, scope, and product coverage for this year's clinical releases

Clinical Solution 2.0 · Limb Sensor Rev K · Aria Scratch Sensor

Decision Overview

The decision concerns the **timing, scope, and product coverage** of Zephyr firmware adoption during this year's **clinical** releases. The Aria Maruho release is intentionally left out of the clinical product line decision because it follows a different regulatory strategy and already runs Zephyr, but the firmware differs significantly from the clinical target. Aria still provides an opportunity to test the clinical Zephyr firmware and can influence the decision below.

Key Releases Impacted

- **Clinical Solution 2.0** — new feature set; scope/timeline TBD (Nov release).
- **Limb Sensor Rev K (stretch goal)** — new HW revision with MFG/verification/regulatory work.
- **Aria Scratch Sensor (target)** — Q3 2026; Zephyr FW feature-complete, needs refinement.

Additional minor releases are planned for clinical as well as other products.

Value & Risk Summary

Value Added by Zephyr

- Platform consolidation, reuse, and tooling gains.
- Forward-looking alignment with vendor roadmap.
- Potential improvements in development speed and quality.

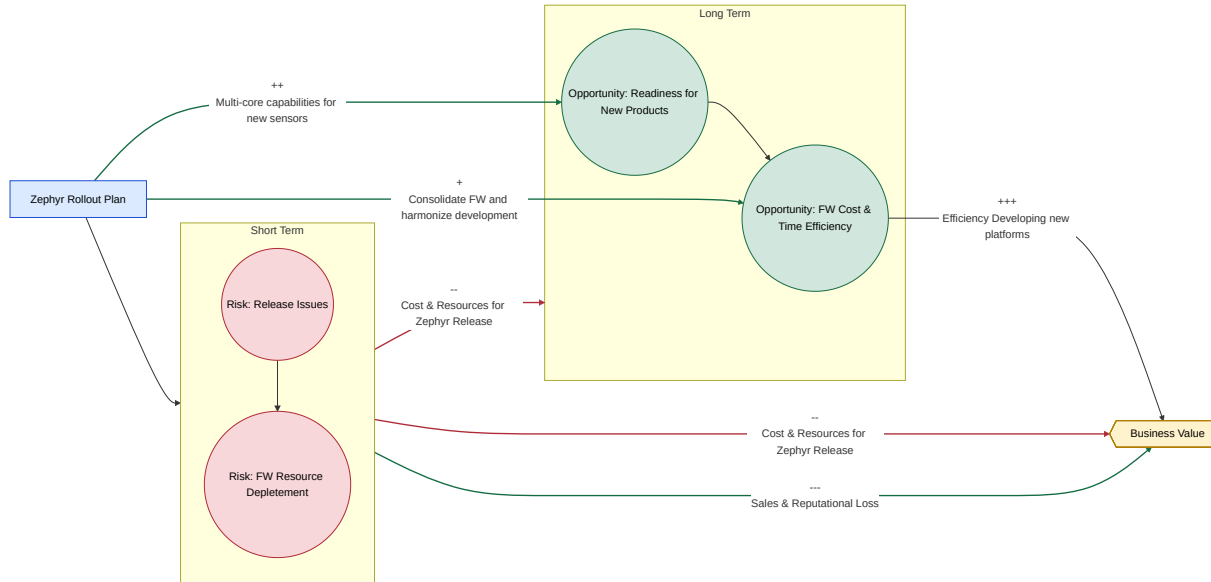
Key Uncertainties

- Regulatory verification may still be per-platform.
- Legacy devices may prevent full harmonization.
- Multi-core SoCs increase complexity and instability risk.

Notable Risks

- Schedule compression (testing/MFG integration excluded).
- Migration/OTA constraints with dual-stack maintenance.
- Team and ecosystem maturity risk.

Influence Diagram



Decision Nodes

Code	Strategy	Description
SAFE_ROLL	Conservative Rollout	No clinical Zephyr rollout this year (clinical release remains on legacy FW). Aria decision is separate.
LKWARM_ROLL	Lukewarm Rollout	Clinical Zephyr rollout on new sensor HW only (Dragonfly).
AGGR_ROLL	Aggressive Rollout	Clinical Zephyr rollout on current clinical line plus new HW (Chest RevK + Limb RevJ/K + Dragonfly as applicable).

Uncertainty Model (Short-Term & Readiness)

Strategy	OK_REL	BAD_REL	Condition	READY	NOT_READY
SAFE_ROLL	1.0	0.0	SAFE_ROLL (no clinical Zephyr)	0.7	0.3
LKWARM_ROLL	0.5	0.5	LKWARM_ROLL + REL_OK	0.8	0.2
AGGR_ROLL	0.3	0.7	LKWARM_ROLL + REL_BAD	0.2	0.8
			AGGR_ROLL + REL_OK	0.8	0.2
			AGGR_ROLL + REL_BAD	0.1	0.9

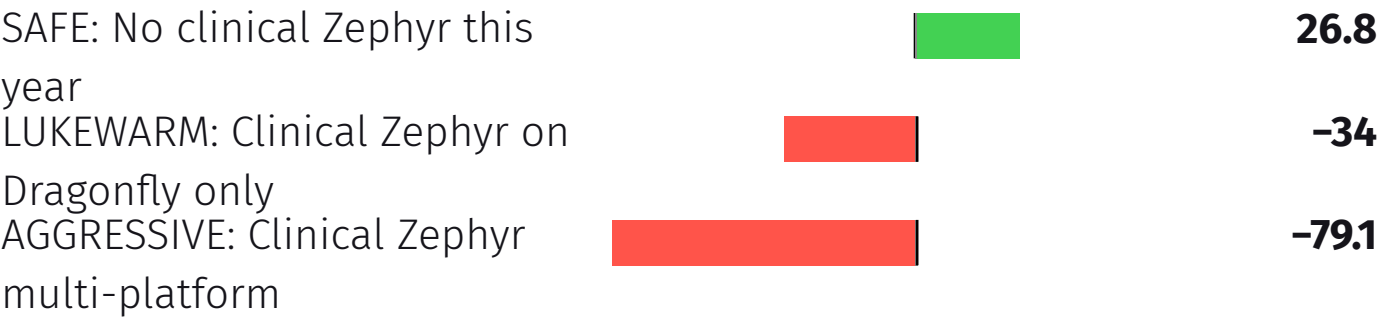
Payoff Model (Summary)

Payoff = Strategy planned cost + (release support cost if BAD_REL) + (long-term readiness value)

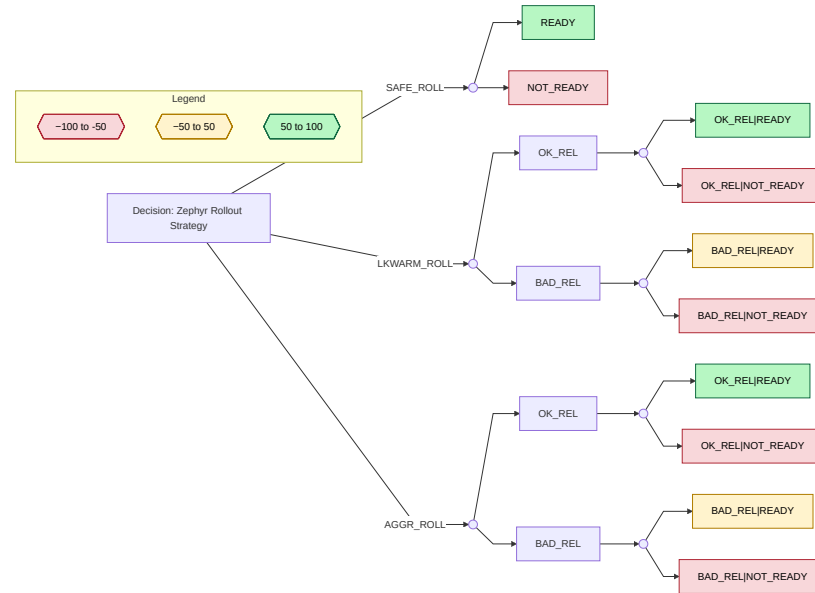
Strategy	Planned Cost	Rationale
SAFE_ROLL	-5	Minimal Zephyr clinical work; preserves capacity for long-term and maintenance.
LKWARM_ROLL	-15	Limited Zephyr clinical scope (Dragonfly only).
AGGR_ROLL	-25	Broad clinical Zephyr scope (largest verification and integration effort).

Expected Value Analysis

Strategy	Expected Value (pts)
AGGRESSIVE: Clinical Zephyr multi-platform	-79.1
LUKEWARM: Clinical Zephyr on Dragonfly only	-34
SAFE: No clinical Zephyr this year	26.8



Decision Tree



Outcome Breakdown (Best Strategy)

Release outcome	Long-term outcome	Prob	Payoff (pts)	EV contribution
No clinical Zephyr release	Zephyr is ready long-term	0.7	49	34.3
No clinical Zephyr release	Zephyr is NOT ready long-term	0.30000000000000004	-25	-7.500000000000001

Total EV: 26.8 points